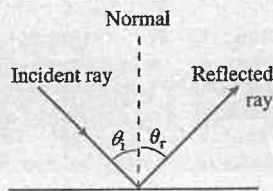


CHAPTER 21 SUMMARY

Reflection and Plane Mirrors

(Section 21.1) Geometrical optics is the approximation that light travels generally in straight lines, called **rays**. **Diffuse reflection** occurs when light rays striking a rough surface are reflected in many different directions. Rays striking a smooth surface like a mirror exhibit **specular reflection**. Light always reflects at an angle equal to the incidence angle.

Law of reflection: $\theta_i = \theta_r$

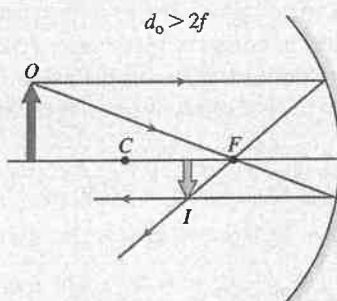


Spherical Mirrors

(Section 21.2) Ray tracing locates images formed by mirrors and lenses. Plane mirrors produce **virtual images**. Spherical mirrors can be either **concave** or **convex**. **Concave mirrors** form real or virtual images, depending on the object distance and focal length. **Convex mirrors** always form virtual images on the other side of the mirror.

Mirror equation: $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

Mirror magnification: $M = \frac{h_i}{h_o} = -\frac{d_i}{d_o}$



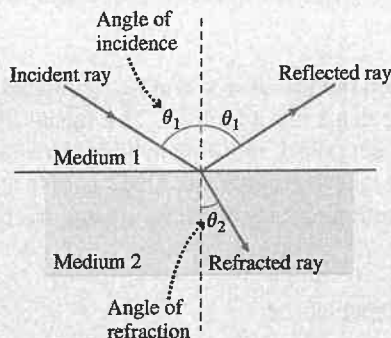
Refraction and Dispersion

(Section 21.3) Light **refracts** when it passes from one medium to another, traveling slower in a medium with a higher **index of refraction**. **Snell's law** describes refraction quantitatively. **Total internal reflection** occurs when light propagating in a medium with lower refractive index is incident on a medium with higher index, provided the incidence angle is greater than the **critical angle**. **Dispersion**, as when white light is sent through a prism, results from different wavelengths of light having different refractive indexes.

Index of refraction: $n = \frac{c}{v}$

Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$

Critical angle: $\theta_c = \sin^{-1}\left(\frac{n_1}{n_2}\right)$

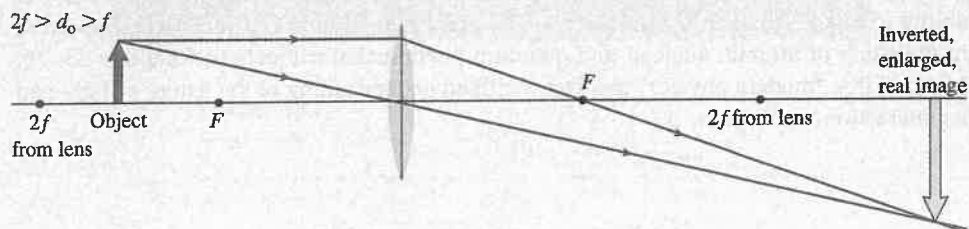


Thin Lenses

(Section 21.4) Convex (converging) lenses form real or virtual images, depending on the object distance and focal length. Parallel rays striking a convex lens are directed through the focal point. For a concave (diverging) lens, parallel rays diverge from the opposite focal point, so concave lenses always form virtual images.

Thin-lens equation: $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$

Lens magnification: $M = -\frac{d_i}{d_o}$



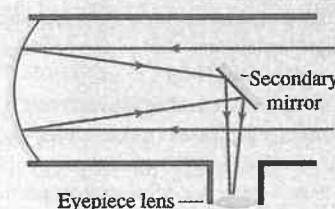
Microscopes and Telescopes

(Section 21.5) A single-lens magnifier uses a convex lens to form a virtual image. **Compound microscopes** and **refracting telescopes** use paired lenses (objective and eyepiece), with the objective forming an intermediate real image and the eyepiece forming a virtual image of the intermediate image. **Reflecting telescopes** replace the objective lens by a parabolic mirror, which allows much greater light-gathering power and eliminates aberrations.

Single-lens magnification (angular magnification): $m = N\left(\frac{1}{f} - \frac{1}{d_i}\right)$

Compound microscope magnification: $M = -\frac{N(L - f_e)}{f_o f_e}$

Telescope magnification (angular magnification): $m = -\frac{f_o}{f_e}$

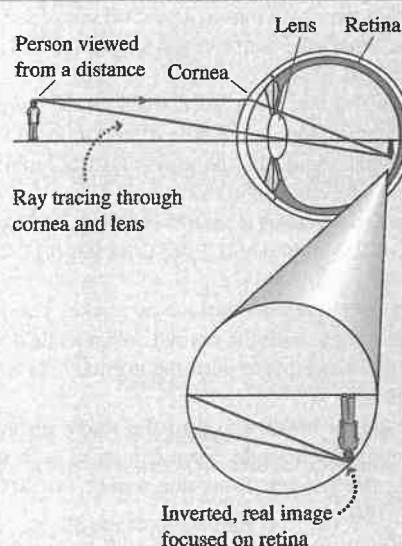


The Eye and Vision

(Section 21.6) In the eye, incoming light passes through the **pupil** and is refracted by the **cornea** and **lens**, which comprise a converging lens system that forms a real image on the **retina**. Common refractive disorders are myopia (nearsightedness), hyperopia (farsightedness), and astigmatism.

Refractive power: $P_{\text{refractive}} = \frac{1}{f}$

Lens-maker's equation: $P_{\text{refractive}} = \frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$



NOTE: Problem difficulty is labeled as ■ straightforward to ■■■ challenging. Problems labeled BIO are of biological or medical interest.

Conceptual Questions

1. Explain the difference between specular and diffuse reflection.
2. If you stood between two parallel vertical mirrors, what would you see?
3. Can a single flat mirror ever produce a real image?
4. Two vertical mirrors are mounted at right angles. Show that this combination allows you to see yourself as others see you.
5. How does a blue wall appear if it's illuminated with blue light? With red light?
6. How can you determine the focal length of a concave mirror? Of a convex mirror?
7. Draw carefully a circular arc to represent the surface of a spherical concave mirror. Drawing a ray diagram with incident rays parallel to the principal axis, show that spherical aberration isn't

- a problem for rays close to the principal axis but becomes more problematic the farther the incoming rays are from the axis.
8. Draw carefully a parabola to represent a section through a parabolic mirror. Drawing a ray diagram, show that all incoming parallel rays focus at the same point.
 9. Why is the order of the colors in the secondary rainbow reversed from the primary rainbow?
 10. A glass fish tank is filled with water. Is it possible for light originating within the water to undergo total internal reflection at the water-glass interface? Can light that travels through the water and glass suffer total internal reflection at the glass-air interface?
 11. Using a convex lens, can you ever get a virtual image with magnification 1? Can you get a virtual image with magnification less than 1?
 12. How would the refractive properties of a glass convex lens change if you used it under water?

13. **BIO Animal eyes.** The lens in the eyes of many animals, including humans, has a refractive index that varies from the center to the edge of the lens—a feature that helps correct for spherical aberration. In order to achieve that correction, should the refractive index be greater at the center or the edge?
14. Does a nearsighted person need positive or negative corrective lenses? Which should be stronger, eyeglasses or contact lenses?
15. After age 40, many people need bifocals. For nearsighted people, the main part of the bifocal lens is negative, with a small portion (usually near the bottom) also negative, but one or two diopters weaker than the main lens. What's the purpose of this design?

Multiple-Choice Problems

16. You have a concave mirror with focal length 25 cm. How far from the mirror should you place an object to get a real image 50 cm from the mirror? (a) 25 cm; (b) 35 cm; (c) 50 cm; (d) 75 cm.
17. An object is 80 cm in front of a concave mirror with 50-cm focal length. The image is magnified by (a) 1.3; (b) 1.7; (c) 2.0; (d) 2.7.
18. An object is 0.92 m from a concave mirror, and a real image forms 0.36 m from the mirror. What's the mirror's focal length? (a) 0.18 m; (b) 0.22 m; (c) 0.26 m; (d) 0.30 m.
19. You have a handheld mirror that gives an enlarged virtual image. If want your face to appear doubled in size when you hold the mirror 30 cm away, what should be the mirror's focal length? (a) 20 cm; (b) 40 cm; (c) 60 cm; (d) 80 cm.
20. The speed of light in a dense glass is measured at 1.56×10^8 m/s. The refractive index of this glass is (a) 1.56; (b) 1.92; (c) 2.24; (d) 3.69.
21. Light propagating in air strikes the surface of a lake. The reflected ray makes a 62° angle with the normal. What angle does the refracted ray in the water make with the normal? (a) 42° ; (b) 48° ; (c) 54° ; (d) 58° .
22. A diver shines a flashlight upward toward the water surface. What's the maximum angle the light beam can make with the surface normal and still emerge from the water? (a) 30.0° ; (b) 41.7° ; (c) 48.8° ; (d) 60.0° .
23. A lightbulb placed 30 cm from a convex lens with focal length 18 cm forms an image magnified by a factor of (a) 0.67; (b) 1.0; (c) 1.5; (d) 1.7.
24. You have a convex lens with focal length 15 cm. In order to form an upright image of newspaper type magnified by a factor of 2.5, how far should you hold the lens from the newspaper? (a) 6 cm; (b) 9 cm; (c) 12 cm; (d) 25 cm.
25. The focal lengths of a telescope's objective and eyepiece lenses are 100 cm and 4.0 cm, respectively. The telescope forms an image with angular magnification (a) 50; (b) 25; (c) 10; (d) 5.
28. ■■ A laser is mounted on a table 1.06 m above the ground, pointed at a vertical mirror a horizontal distance 3.50 m away. The laser beam strikes the mirror 1.62 m above the ground. Behind the laser is a wall, 2.80 m from the aperture where the laser light emerges. At what height does the reflected beam strike the wall?
29. ■■ A dog sitting 2.0 m from a vertical mirror thinks its image is another dog. (a) How far is the image from the dog? (b) To investigate, the dog walks directly toward the mirror at 0.25 m/s. At what rate does the dog approach its image?
30. ■ Figure P21.30 shows two mirrors at right angles. Light strikes one mirror at 45° , as shown. (a) Continue the ray diagram and show that after striking both mirrors the outgoing ray is parallel to the incoming ray. (b) What is the distance between the incoming and outgoing rays?
31. ■■ Repeat the preceding problem for an incidence angle of 30° .
32. ■■ You're standing on the dock, holding a flashlight 3.0 m above the surface of a lake. A boat is 20 m away. You want to shine your flashlight so the light reflected from the water strikes the boat's bow, 1.2 m above the water line. Where should you aim the flashlight?

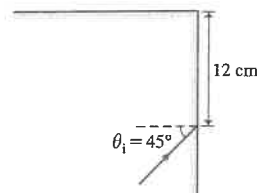


FIGURE P21.30

Section 21.2 Spherical Mirrors

33. ■ A concave mirror has focal length 35 cm. Where should you place an object to get a real image (a) 50 cm from the mirror and (b) 1.0 m from the mirror?
34. ■ An object is 66.0 cm from a concave mirror, and a real image forms 48.0 cm from the mirror. Find (a) the mirror's focal length and (b) the image magnification.
35. ■■ Sketch ray diagrams, including object and image, for a concave mirror when (a) $M = -1$; (b) $M = -0.5$; (c) $M = -3$; (d) $M = +1.5$.
36. ■ (a) Draw the ray diagram for a concave mirror with an object 100 cm from the mirror and an inverted image 25 cm from the mirror. (b) What's the mirror's focal length?
37. ■■ Draw the ray diagram for a convex mirror with an object 100 cm from the mirror and an upright image 25 cm on the other side of the mirror.
38. ■ You place a small lightbulb 70 cm from a concave mirror and find a real image 20 cm from the mirror. Where should you place the bulb so its real image is 70 cm from the mirror?
39. ■ An object is 125 cm in front of a concave mirror with focal length 75.0 cm. (a) Where is the image? (b) Is it real or virtual? (c) What's the magnification?
40. ■ A small lightbulb 2.0 m from a concave mirror gives a real image 6.0 cm from the mirror. (a) Find the mirror's focal length. (b) Where will the image be if the bulb is moved to 4.0 m from the mirror?
41. ■ A small lightbulb is 35 cm in front of a concave mirror with focal length 21 cm. What are the location, magnification, and orientation (upright or inverted) of the bulb's image?
42. ■■ You have a handheld mirror that gives a virtual image of your face, enlarged by a factor of 1.5, when you hold the mirror 30 cm away. What's the mirror's focal length?

Problems

Section 21.1 Reflection and Plane Mirrors

26. ■ A 1.60-m-tall student stands before a vertical mirror. She can just see her whole body, from head to toe. (a) How tall is the mirror? (b) If the student's eyes are 10 cm below the top of her head, how far is the bottom of the mirror above the ground?
27. ■■ You're standing 3.0 m from a plane mirror. If you want to photograph your image, at what distance should you focus the camera's lens?

43. ■■ A concave mirror has focal length 50 cm. You want this mirror to form a real image 1.2 times larger than an object. (a) Where should you place the object? (b) Draw a ray diagram to locate the image. (c) Where should you place the same object so there's a virtual image 1.2 times larger than the object?
44. B10 ■■ **Dentist's mirror.** A dentist's mirror produces an upright image of a tooth, magnified three times. (a) Is the mirror concave or convex? (b) If the dentist holds the mirror 2.0 cm from the tooth to achieve this magnification, what's the mirror's focal length?
45. ■■ Looking into a 6.0-cm-diameter reflective ball, you see your face half size. (a) Is your image upright or inverted? (b) Where is the image, relative to the center of the ball? (c) How far is your face from the surface of the ball?
46. ■■ A car's outside rear-view mirror is convex, with focal length -1.0 m. In the mirror you see a truck that's actually 3.5 m tall and 10.0 m behind you. What are its apparent height and location?
47. ■■ A spherical mirror is silvered on both sides, so it's concave on one side and convex on the other. When an object is 60 cm from one side, a real image forms 30 cm from that side. (a) Find the curvature radius. (b) Describe the image if the same object is held 60 cm from the other side of the mirror.
48. ■■■ Consider the illustrations in Figures 21.7a and 21.7b. Prove using geometry that reflected rays reach the focal point $f = R/2$ in the limit as the incoming rays approach the principal axis. *Hint:* Consider the triangle formed by the radius of curvature, principal axis, and reflected ray, and use the law of sines.
57. ■■ You're standing in waist-deep water with your eyes 0.52 m above the surface. A fish is swimming 0.65 m below the surface. Your line of sight to the fish makes a 45° angle with the water's surface. How far (horizontally) from you is the fish?
58. ■■ Looking directly downward from a boat, you see a shark that appears to be 5.0 m below the water's surface. What's the shark's actual depth?
59. ■■ What's the critical angle for light going from diamond into air? How does this value help explain diamond's appearance?
60. ■■ Find the speed of violet light ($\lambda = 400$ nm) and red light ($\lambda = 700$ nm) in silicate flint glass. By what percentage do the speeds differ?
61. ■■■ Light in air strikes a glass pane with $n = 1.50$, making a 35° angle with respect to the normal. (a) What angle does the light make with the normal in the glass? (b) The glass is 2.0 mm thick. As shown in Example 21.5, the ray emerging from the glass is parallel to the ray that entered the glass. Find the perpendicular distance between those parallel rays.
62. ■■■ A prism is in the shape of an equilateral triangle. An incoming beam of white light strikes the prism at 78° with respect to the normal, as shown in Figure P21.62. The refractive indices for far red (700 nm) and far violet (400 nm) are 1.62 and 1.65, respectively. Find the angles at which the two colors emerge on the other side of the prism, and compute the angular spread of the light. *Note:* For simplicity, Figure P21.62 shows a single ray of one color.

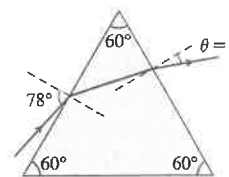


FIGURE P21.62

Section 21.3 Refraction and Dispersion

49. ■ For which material in Table 21.2 does light travel most slowly? What is that speed?
50. ■ The speed of light in a new type of plastic is measured at 2.11×10^8 m/s. What is its refractive index?
51. ■ Find the speed of yellow light in (a) ice; (b) quartz; (c) ethanol; (d) diamond.
52. ■ A park ranger's flashlight beam strikes a lake surface. The reflected ray makes a 55° angle with the normal. What angle does the refracted ray make with the normal in the water?
53. ■■ A rectangular glass block ($n = 1.53$) lies submerged at the bottom of a tank of water (Figure P21.53). A light beam enters the water from the air above, making a 20° angle to the normal in air. Determine the angle the light ray makes with the normal in (a) the water and (b) the glass.

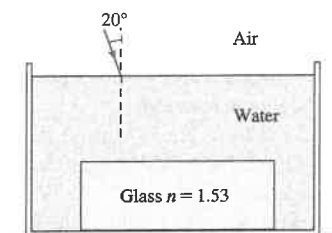


FIGURE P21.53

54. ■■ You're looking straight down into a 2.35-m-deep swimming pool. What's the apparent depth?
55. ■■ In winter a lake has a 0.25-m-thick ice layer over 1.10 m of water. A light beam from above strikes a spot on the ice, making a 30° angle with the normal. How far horizontally from that spot does the light ray strike the lake bottom?
56. ■■ A diver shines a flashlight upward toward the water surface. Describe what happens when the flashlight beam strikes the

water's surface with these angles, measured from the normal: (a) 20° ; (b) 40° ; (c) 60° .

57. ■■ You're standing in waist-deep water with your eyes 0.52 m above the surface. A fish is swimming 0.65 m below the surface. Your line of sight to the fish makes a 45° angle with the water's surface. How far (horizontally) from you is the fish?
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Section 21.4 Thin Lenses

63. ■ For a convex lens with focal length 50 cm, draw ray diagrams to show image formation for an object (a) 150 cm; (b) 80 cm; (c) 35 cm from the lens.
64. ■ Repeat the preceding problem for a concave lens with $f = -50$ cm.
65. ■ For a convex lens draw ray diagrams for the following cases: (a) $M = -0.5$; (b) $M = -1.0$; (c) $M = -2.0$.
66. ■■ A convex lens forms an image of a 12-cm-tall soda can. The image is 2.0 cm tall and appears 8.2 cm from the lens. (a) Where is the can? (b) What is the lens's focal length?
67. ■■ You're using a 20-cm-focal-length convex lens to image a small lightbulb. (a) Where should you place the bulb to get a half-size real image? Where is that image? (b) Where should you place the bulb to get a double-size real image? Where is that image?
68. ■■ A small lightbulb is 1.00 m from a screen. If you have a convex lens with 20-cm focal length, where are the two lens locations that will project an image of the lightbulb onto the screen? What's the magnification in each case?
69. ■■ A movie projector uses a single lens to project a real image on a screen 6.0 m from the lens. Each frame of the movie film is 3.0 cm tall, and the image is 1.20 m tall. (a) Should the lens be concave or convex? (b) Should the film be upright or inverted in the projector? (c) How far should the film be from the lens?
70. ■■ You're using a concave lens with $f = -5.4$ cm to read 4.0-mm-high newspaper type. How high do the type characters appear if you hold the lens (a) 1 cm; (b) 3 cm; (c) 10 cm from the newspaper?

71. ■■ Repeat the preceding problem for a convex lens with $f = +5.4$ cm, considering lens-to-newspaper distances (a) 1 cm; (b) 3 cm; (c) 5 cm.
72. ■■■ A small lightbulb is fixed to a meterstick's 0-cm mark, and at the 100-cm mark is a concave mirror with focal length 10 cm. At the 50-cm mark is a convex lens with a focal length of 20 cm. Describe the final image formed by the mirror.
73. ■■ You're using a convex lens with $f = 11.0$ cm to study an insect. How far from the insect should you hold the lens to get an upright image magnified by a factor of 1.8?
74. ■■■ You're using a 16-cm-focal-length convex lens as a magnifying glass, with $M = +3.0$. How far and in what direction (relative to the object) must you move the lens to change the magnification to -3.0 ?

Section 21.5 Microscopes and Telescopes

75. ■ A 4.5-cm-focal-length lens is used as a magnifying glass. (a) What's the maximum angular magnification possible for a person with near point at 25 cm? (b) Repeat for an older person with $N = 75$ cm.
76. ■ A magnifying glass has a maximum angular magnification 10.2, for a person with near point at 30 cm. What's the lens's focal length?
77. ■■ A person with near point at 25 cm views a 0.75-mm-long flea. (a) What is the flea's angular size (a) when viewed at the near point and (b) when viewed through a magnifying glass that enlarges the image 10 times?
78. ■■ A compound microscope has a 0.85-cm-focal-length objective and a 1.45-cm focal length eyepiece. (a) If the lenses are 11 cm apart, what's the microscope's magnification? Assume a 25-cm near point. (b) What should be the focal length of a new eyepiece to increase the magnification by 20%?
79. ■■ A compound microscope has magnification $M = -125$ and a 1.50-cm-focal-length objective located 11.7 cm from the eyepiece. The user has a 30-cm near point. What's the eyepiece's focal length?
80. ■ A telescope uses two convex lenses: an objective with $f = 120$ cm and an eyepiece with $f = 6.0$ cm. (a) What's the angular magnification? (b) Is the image upright or inverted?
81. ■■ Draw the ray diagram for the Galilean telescope, showing that the final image is upright.
82. ■■ (a) Explain how you would make a telescope from two convex lenses with focal lengths 1.2 cm and 24 cm. (b) What would be the telescope's angular magnification?
83. ■■■ (a) Explain why the length of a telescope using two convex lenses should be approximately equal to the sum of the lenses' focal lengths. (b) A telescope with a magnification of -40 has two lenses separated by 95 cm. Find the focal lengths of the two lenses.

Section 21.6 The Eye and Vision

84. ■ A convex lens ($n = 1.52$) has curvature radius 30 cm on both sides. (a) Find its focal length and refractive power. (b) Repeat for a lens that's concave, with 30-cm curvature on both sides.
85. ■■ A thin lens ($n = 1.58$) is convex on both sides, with curvature radii 40 cm on one side and 20 cm on the other. (a) Find the focal length. (b) Why doesn't your answer depend on which side faces the incoming light?
86. ■■ Draw a diagram like that in Figure 21.41 for a lens that's concave on both sides. Explain how the sign convention leads to a negative focal length and negative refractive power.
87. If a person's far point is at 39 cm, what should be the diopter power of corrective eyeglasses? Assume that the eyeglasses are 1.5 cm from the eye.
88. ■■ A patient is prescribed -4.5 -diopter contact lenses. Where is this patient's far point?
89. ■■ Professor Deadwood has to hold a book at arm's length (65 cm from his eye) to read clearly. Find the refractive power of glasses that will reduce that distance to a more comfortable 30 cm.
90. ■■ A $+2.75$ -diopter contact lens is prescribed for a patient, letting her see clearly objects as close as 25 cm. Where is her near point?
91. ■■■ A **plano-convex** lens is flat on one side and convex on the other. (a) Find a general expression for the refractive power of a plano-convex lens in terms of its refractive index n and curvature radius R on the convex side. (b) If $n = 1.65$, find the curvature radius needed for a $+5.0$ -diopter lens.
92. ■■■ A **meniscus** lens is concave on one side and convex on the other. (a) Find the refractive power of a meniscus lens with $n = 1.60$ and curvature radius 30 cm on the convex side and 40 cm on the concave side. (b) Repeat for a similar lens but with the two curvature radii reversed.

General Problems

93. ■■■ Optical scientists are developing materials whose refractive indices change when an electric field is applied. One application may be zoom lenses in cell phone cameras. Suppose such a lens initially has index n_1 and focal length f_1 . Find an expression for the refractive index n_2 required to triple the focal length.
94. ■■ By what factor is the image magnified when an object is placed 1.5 focal lengths from a converging lens? Is the image upright or inverted?
95. ■■ A lightbulb is 56 cm from a convex lens, and its image appears on a screen 31 cm on the other side of the lens. (a) What's the lens's focal length? (b) By how much is the image enlarged or reduced?
96. ■■■ **Squid vision.** The giant squid's dinner-plate size eye is the largest in the animal kingdom, and boasts a lens some 12 cm in diameter. That lens size, along with a light-emitting organ just below the eye, allows the squid to see in the dim ocean depths. If the retina lies 27 cm behind the lens, and the lens has refractive index 1.52, what should be the lens's curvature radius (assuming it has the same magnitude on both sides) to focus distant objects? Assume the lens is surrounded by fluid with water's refractive index—1.33—on both sides, and use a generalization of the lens-maker's equation for the case when the surrounding medium isn't air:
$$P_{\text{refractive}} = \frac{1}{f} = \left(\frac{n_{\text{lens}}}{n_{\text{surroundings}}} - 1 \right) \left(\frac{1}{R_1} - \frac{1}{R_2} \right).$$
 (Actually, the lens's refractive index varies from the center to the edge of the lens—which the squid has in common with many other animals.)
97. ■■ An object is 6.0 cm from the surface of a reflecting ball, and its image appears three-quarter size. Find (a) the ball's diameter and (b) the image location.
98. ■■ A 3.0-cm-long pin is viewed through a concave lens with focal length -6.2 cm. (a) Where should you place the lens so the pin's image is 1.8 cm long? (b) Where is the image located? (c) Draw a ray diagram for this situation.
99. ■■ A candle is 84 cm from a screen. A converging lens placed between candle and screen produces a sharp image on the screen when the lens is in either of two positions. In one position, the magnification is 2 and in the other it's $1/2$. (a) Find the distance from lens to candle in each case. (b) Identify each image as either virtual or real. (c) What is the lens's focal length?
100. ■■ A 300-power compound microscope has a 4.5-mm-focal-length objective lens. If the distance from eyepiece to objective is 10 cm, what should be the focal length of the eyepiece?

101. ■■■ A camera can normally focus as close as 60 cm, but additional lenses mounted in front of the main lens can provide close-up capability. What type and power of auxiliary lens will allow the camera to focus as close as 20 cm? The distance between the lenses is negligible.
102. ■■■ Two vertical plane mirrors are placed with a 72° angle between them. You stand a nail upright, on the line bisecting the mirrors. How many images of the nail appear? Draw a ray diagram showing the image formation.
103. B10 ■■■ **Contact lenses.** A contact lens is in the shape of a convex meniscus (see Problem 92). The inner surface is curved to fit the eye, with curvature radius 7.80 mm. The lens is made from plastic with $n = 1.56$. If the lens is to have a focal length of 44.4 cm, what should be the curvature radius of its outer surface?

Answers to Chapter Questions

Answer to Chapter-Opening Question

Laser surgery reshapes the cornea, changing its refractive power in order to focus images on the eye's retina.

Answers to GOT IT? Questions

Section 21.2 (c) Enlarged, inverted, real

Section 21.3 (b) Bends away from the normal

Section 21.4 (b) $2f > d_o > f$

Section 21.5 (d) 2 cm